

REVIEW

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A memory fabric for conversational AI agents enabling shared and persistent multiuser memory

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Abstract

Conversational artificial intelligence is now the most widely adopted platform for interfacing with large language models. Alongside large language models these artificial intelligence systems rely on contexts derived from past conversations and preferences to provide accurate and the most relevant responses to users. The knowledge base and past experiences contribute to long-term memory, while processing ongoing conversations generates short-term memory. Both long-term and short-term memories together provide a comprehensive and coherent context to the user. While most architectures focus on a single user context, there is an emerging need in conversational artificial intelligence to provide a system to generate context from multiple individuals and/or agents. Building on this foundation, we introduce memory fabric, a framework that allows conversational artificial intelligence to leverage context drawn from multiple users to generate coherent responses in a multiuser setting. This review is a synthesis of foundational memory architectures, from early models such as Neural Turing Machines and Key-Value Memory Networks to recent systems like AutoGen, Shared Recurrent Memory Transformer, AgentVerse, and Secured Agent Memory Exchange Protocol evaluated through the lens of memory fabric. It also addresses work on dialogue and agent memory, multi-agent shared-memory designs, privacy and provenance strategies, benchmarking standards, and system-level considerations. A systematic review of english articles using Institute of Electrical and Electronics Engineers Xplore, Association of Computing Machinery, arXiv, SpringerLink, and Association of Computational Linguistics anthology databases was performed. Only full-text studies reported between 2014 and 2025 were included for analysis. Following Preferred Reporting Items for System Reviews and Meta-Analyses guidelines, a total of 662 studies were identified, of which 27 studies met the inclusion criteria. Overall, this review presents a detailed analysis on leveraging contexts from multiple users in a shared environment in foundational memory architectures, offering an in-depth analysis of their strengths and shortcomings, along with recommendations and future research directions.

Keywords Conversational artificial intelligence, Shared memory, Large language models, Copilot, Multi-agent systems, Memory fabric, Generative agents, Long-term memory, Personalization



1 Introduction

Conversational artificial intelligence (AI) has been defined as “the study of techniques for creating software agents that can engage in natural conversational interactions with humans” [1]. Conversational AI enables the design of AI-powered conversational agents that are “software systems that simulate interactions with real people” by engaging through text, speech, gestures, and other body expressions [2]. Conversational agents have found widespread application across various domains, including retail [3], banking [4], education [5], hospitality and tourism [6], healthcare [7] and media and entertainment [8].

AI agents are transforming the technology industry by providing an interface to expertly leverage trained Large Language Models (LLMs) to perform a task [9]. AI agents are autonomous software systems that perform tasks by sensing their environment, reasoning about it, and executing actions to achieve task-specific goals. Adoption of AI agents is growing rapidly across all kinds of industries and applications. There are several types of AI agents, including generative agents, learning agents, goal-oriented agents & others. Generative AI agents, due to their ability to generate human-like content, have been the most widely used throughout the world. Conversational agents such as ChatGPT, Google Assistant etc. are types of generative AI agent that interact with humans using natural language. These agents use context and dialogue together with instructions to answer questions. To accurately answer questions with personalization, these agents persistently store the conversation history, context, preferences, and other information. Most of the existing studies focus on per-user memory, where a single AI agent supports an individual user. However, this approach has several limitations, including the lack of shared context across users, inadequate support for multi-user environments, and scalability challenges. As AI agents are increasingly adopted across diverse applications, there is a strong need for shared, multi-user memory systems to concurrently support multiple users.

To address this gap, this paper introduces “Memory fabric” framework that will enable memories from multiple parties to be combined, exposed, and reconciled. This study aims to understand How can conversational AI systems integrate and govern long term memory in a shared environment using existing architectures. Unlike reviews that are based on subjective judgments of experts, this review uses industry wide benchmarks to evaluate how multi-user shared memory works for existing architectures.

1.1 Foundational memory architectures

This section reviews foundational memory architectures that form the basis of modern conversational AI systems evaluated through the lens of proposed multiuser conversational AI framework Memory Fabric.

Neural Turing Machines (NTM): Graves et al. introduced differentiable, addressable external memory that a neural controller can read/write, enabling algorithmic like behavior by combining computation with memory [11]. NTMs were groundbreaking in the field of neural networks. They provide evidence that neural

networks can learn predictable algorithmic behavior. Most widely used neural networks today are memory augmented, which is derived from the idea behind NTMs. For instance, neural networks before NTMs would store the key-value pairs in their hidden state for simple storage and retrieval. NTMs demonstrated that it is possible to write the

data into external memory and read from it directly using a controller, thereby working on a higher scale with a reusable algorithm.

Differentiable Neural Computer (DNC): Often referred to as NTM 2.0, DNCs are a more advanced version of NTM. DNCs do better memory management by tracking free memory slots instead of always overwriting. DNC also stores the order of writes to memory, which allows it to retrace the sequences, which allow it to solve problems such as tree or graph traversal [12].

Memory Networks / End-to-End Memory Networks / Key-Value Memory Networks: These are a family of architectures that show how external memory together with attention improves QA and retrieval tasks. For instance, key-value memory network stores a key used for addressing and the value as the content encoding, which is widely reused in retrieval-augmented systems. These networks can perform multi-hop reasoning to allow chaining different tasks together [13].

Dynamic Memory Networks (DMN): This network reasons over the input using the current memory and the question in context. The process is repeated multiple times until reasoning is complete and answers the question. This dynamic evolution

of memory is referred to as the episodic memory module. Compared to the key-value memory, DMN are more flexible as it processes multiple types of texts while reasoning instead of a fixed type of lookup [14]. Figure 1 shows the evolution of memory-augmented architectures spanning from 2014–2025.

These foundational architectures show the two recurring building blocks used later: (1) an addressable store (vector, KV, graph), and (2) an adaptive read/write/update policy.

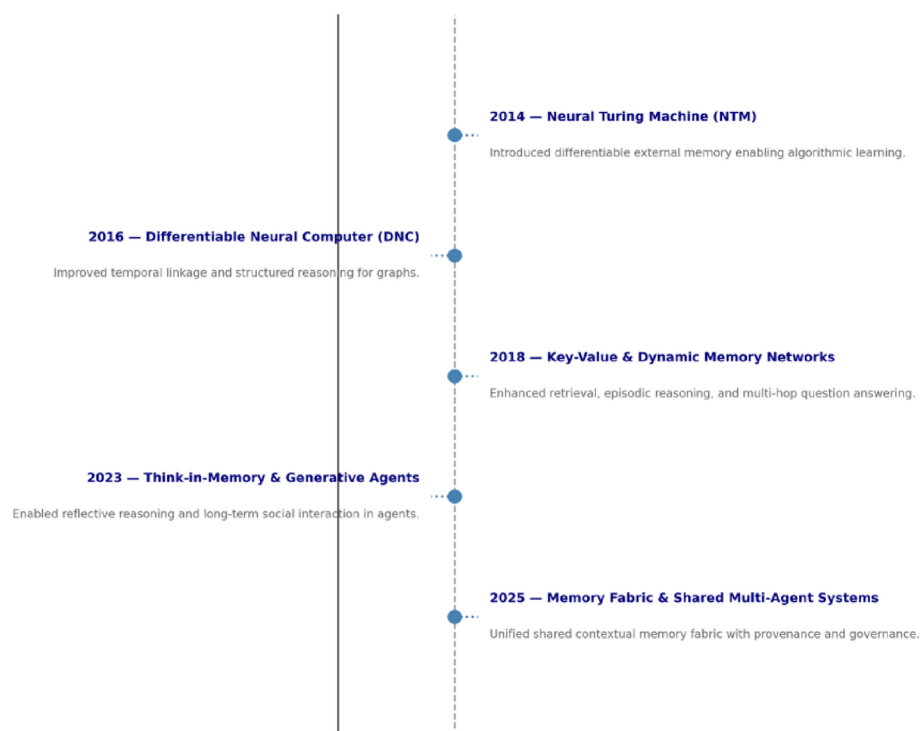


Fig. 1 Evolution of memory-augmented architectures (2014–2025), showing the progression from Neural Turing Machines and differentiable memory models to modern agent-based and long-context foundation model frameworks

These memory augmented neural networks create embeddings of individual users only assuming a single logical owner for both short- & long-term memories.

In this paper, we have examined the foundational memory architectures and evaluated them through the lens of the proposed multiuser-shared conversational memory architecture Memory fabric. Multiuser-shared conversational memory can be divided into the following categories based on the literature reviewed as shown in Fig. 2:

1. Tiered memory architectures - systems that keep per-user private memories and connect them to one or more shared/team-level memories with controlled bridges.
2. Centralized shared memory with role-based policies - a central repository with RBAC/permission graphs determining visibility.
3. Privacy-preserving shared fabrics - memory remains local but is selectively revealed or aggregated using DP/encryption or protocols.
4. Multi-agent memory ecosystems - each agent holds private memories but can share them with common channels (often in agent simulation frameworks).
5. Graph-structured memory + provenance - systems where memory is stored as nodes/edges with provenance metadata enable merging, conflict detection, and reasoning.

This review is organized according to the following sections: Sect. 2 describes the methods used to perform this review including data collection, data analysis etc.. Sect. 3 presents the findings of our analysis. Sect. 4 presents results of benchmarking across multiple dimensions of memory fabric. Section 5 discusses our principal findings, outlines various approaches for implementing the memory fabric, and highlights its advantages, disadvantages, and future research directions. Finally, section 6 concludes the paper.

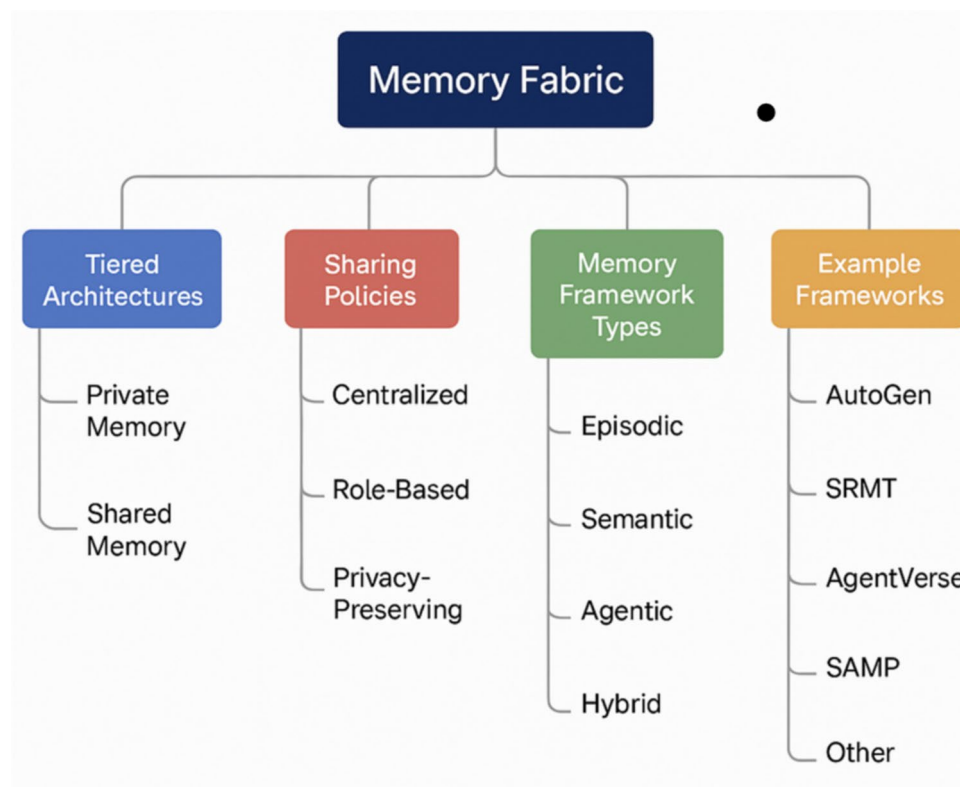


Fig. 2 Key subsystems of the memory fabric framework supporting shared, persistent, and context-aware memory across multiple users

2 Methods

2.1 Search strategy

This review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [10] with a goal to identify, evaluate, and synthesize existing literature on *long-term and shared memory architectures in conversational AI* focusing on frameworks for multi-user and multi-agent memory sharing. A comprehensive search was conducted across Institute of Electrical and Electronics Engineers (IEEE) Xplore, Association of Computing Machinery (ACM) Digital Library, arXiv, SpringerLink, and the Association of Computational Linguistics (ACL) Anthology, covering literature published between January 2014 and May 2025. The chosen time range encompasses all the groundbreaking research in the field of conversational AI and LLMs. Additional studies were identified by examining the reference lists of retrieved papers. Figure 3 illustrates the workflow for the systematic identification of relevant literature. The search strategy employed combinations of keywords related to “conversational AI” and “memory-augmented architectures,” linked using the Boolean operators “OR” (to combine terms within a domain) and “AND” (to connect terms across domains). The full set of terms used to retrieve primary studies is provided in Table 1.

2.2 Search result selection criteria

The computational architectures retrieved from running the query stated in Table 1 across the databases was analyzed with respect to persistent, long term or shared memory by conversational agents. The initial search yielded 412 studies from the selected databases and an additional 250 studies from supplementary sources, including reference lists and citation chaining. These 662 records were compiled in a spreadsheet for deduplication. A total of 342 duplicates were identified and removed, leaving 320 unique studies for further analysis, as shown in Fig. 3 (screening section). The remaining 320 studies underwent title and abstract screening to identify work related to multi-user interactions and privacy mechanisms in multi-agent environments. This step resulted in 70 studies eligible for full-text review, as indicated in the eligibility section of Fig. 3. Full texts of these 70 studies were then evaluated against predefined inclusion and exclusion criteria (summarized in Fig. 3) to determine their suitability as primary studies.

Studies were included if they: (1) focused on implemented memory and persistent or shared memory; (2) were written and published in English; (3) published between December 2014 and December 2025; (4) provided full-text access; (5) presented original research published in journals, conference proceedings, or preprints; and (6) focused on multi-user settings. Studies were excluded if they: (1) were review papers rather than primary research; (2) did not pertain to long-term memory in conversational AI; (3) lacked full-text availability; (4) were published before 2014. Finally, After applying these criteria, 27 studies were selected for detailed analysis, as shown in the included phase of Fig. 3.

2.3 Data extraction and evaluation

With the 27 articles identified, we conducted a more detailed review of the shortlisted studies. The data were extracted from all studies meeting our inclusion criteria for the review. Each of these studies use a structured format beginning with architecture and memory type from NTM, DNC to Shared Recurrent Memory Transformer (SRMT)

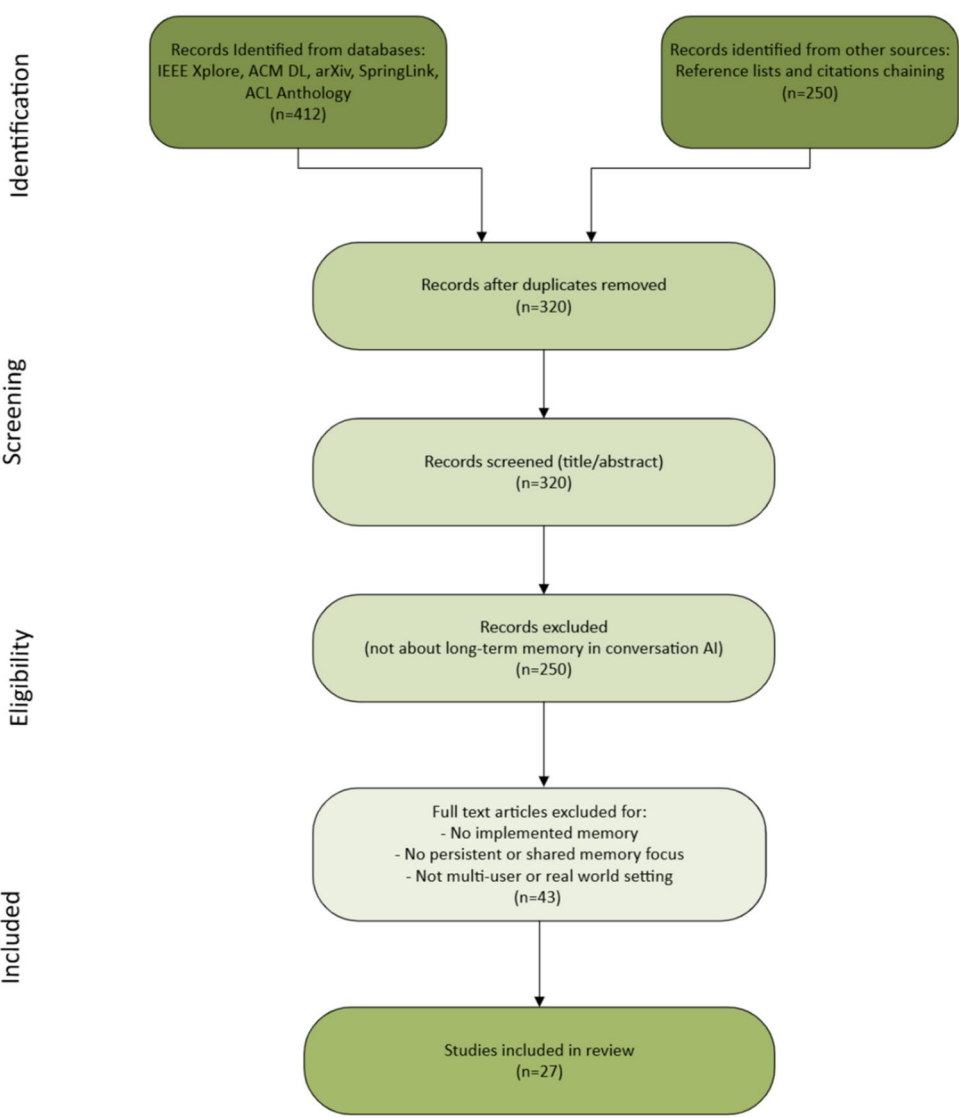


Fig. 3 PRISMA workflow for systematic identification of scientific literature

Table 1 Search query for the study retrieval

Search query
("memory-augmented neural network" OR "neural turing machine" OR "differentiable neural computer" OR "key-value memory network" OR "memory fabric" OR "multi-agent memory" OR "generative agents" OR "memory sharing" OR "shared memory") AND ("conversational AI" OR "dialogue system" OR "retrieval-augmented generation" OR "long-term memory" OR "multi-user memory")

evaluating each on memory policy such as episodic vs persistent storage. As we are dealing with shared memory, data about privacy features such as access control and provenance that are offered by these architectures is also extracted. Finally, existing benchmarking frameworks were used to validate short- and long-term memories in multiuser memory fabric. Through this process, we aim to highlight the most widely used memory-augmented foundation architectures, emphasize their advantages, address the challenges, and identify potential areas for innovation and improvement. The data for all studies were independently extracted by all authors (VG, AT), and any discrepancies

were resolved through mutual discussion between the authors. The extracted data were finally evaluated by all authors independently and consensus was reached through mutual agreement.

3 Results

3.1 Foundational memory architecture review

The foundational work in memory-augmented neural architecture laid the foundation for memory fabric in AI. Graves et al. (2014) introduced Neural Turing Machines (NTMs), which combined end-to-end differentiable controllers with an external memory matrix, enabling read and write operations via differentiable attention mechanisms. Their experiments demonstrated that NTMs could learn simple algorithms such as copying and sorting, showing the feasibility of learned read/write operations. While NTMs provided a low-level abstraction for shared memory, they lacked access control, unlike conventional computer systems that support permissions such as read-only, write-only. NTMs also had no support for provenance, where it is not possible to track history of memory such as version history. Together these limitations prevented direct application in multi-user systems.

Building on this, Graves et al. (2016) proposed the Differentiable Neural Computer (DNC), which enhanced memory addressing and management algorithms by applying structured reasoning tasks using graphs. DNCs demonstrated the ability to store and traverse graph-like structures using learned addressing. These capabilities made DNCs particularly suitable for graph-like structures of shared memories in a multi-user context. Complementary approaches, such as Memory Networks introduced architectures combining long-term memory modules with learned attention-based reading mechanisms, showing that memory-attention patterns are highly effective for retrieval and reasoning. Many modern retrieval-augmented based generative agents have been influenced by these patterns. In context of memory fabric, applications using such architecture require a role-based access policy to control access to memories.

Key-Value Memory Networks [13] further refined memory designs by separating key encoding for addressing and value encoding for output like a hash-map or dictionary. Their experiments on document Q&A presented improved performance in retrieval and output accuracy. Key-Value memory networks can naturally be applied to shared memory scenarios where keys are created by using a key encoder on user metadata and the raw data, while the value encoder stores the content. Thus, allowing multi-hop keywords in the context of more than one user. Dynamic Memory Networks (DMNs) introduced episodic memory formation and iterative attention mechanisms, which enhanced reasoning on compositional tasks such as question answering and visual question answering [14]. DMN works in episodes wherein every episode, each input fact is processed and stores/updates current memory based on the question. This process is repeated until the final memory has the answer. Episodic memory designs perform step-by-step reasoning instead of trying collapse all reasoning in one hence are particularly relevant in multi-user settings, as they enable session-level memory tracking as well as maintaining list of contributions.

More recently, Think-in-Memory proposed storage of intermediate thoughts or post-processing reasoning states to reduce redundant inference, enhancing stability across repeated queries [15]. While TiM demonstrates the benefit of storing intermediate

reasoning artifacts within memory the fabric paradigm, it does not yet incorporate explicit access policies for multi-user environments.

3.2 Dialogue, personalization, and single-user long-term memory

Advances in dialogue personalization and long-term single-user memory provide insights into shared memory models in multi-agent systems. Li et al. (2016) developed persona-based neural conversation models with a notion of speakers' style, identity and consistency [18]. Each speaker is assigned a vector embedding that represents their persona. These are learned together with other aspects of the model during training. Once the speaker embedding is injected into the decoder, the responses are more stable and consistent over conversations. Persona-grounded datasets such as PERSONA-CHAT further showed that explicit persona information increases engagement in dialogue [17].

Meta-learning approaches [19] enable rapid adaptation to new persona data, raising questions in multi-user systems about whether adaptations should remain private or be shared. This balancing between personalization and knowledge sharing raises the governance challenges that memory fabric must eventually address. Memory-Augmented Dialogue Management frameworks introduced slot-value memory structures and external memory to track task-oriented dialogue histories beyond single sessions, demonstrating patterns that are relevant to shared workflows [20]. By maintaining structured memories of user goals, actions, and dialogues, MAD demonstrated how explicit memory can support continuity that can be extended over multiple interactions.

Reflective Memory Management extends these ideas further by coupling dynamic summarization with reinforcement learning-based retrieval refinement [21]. RMM illustrates the importance of controlling memory granularity, deciding not only what information is retained but also at what level of detail. In multi-user settings, this becomes critical for determining which aspects of a memory fabric should be visible to others, which should remain private, and how evolving summaries shape the collective understanding. Benchmarks such as MADial-Bench provide practical evaluation protocols for memory recall and retrieval tasks, essential for measuring the correctness and privacy of multi-user memory systems [22].

3.3 Generative agents, multi-agent memory, and shared memory systems

Generative agents have made a significant step toward mimicking human-like reasoning and social interactions using multi-agent memory. Park et al. (2023) introduced a system of generative agents with a persistent memory composed of short summaries of conversations and experiences [16]. The agents could plan future actions based on memories and reflective reasoning in a test environment. The study demonstrated that combining short term memory storage, retrieval, and reflection enables emergent behaviors such as understanding long-term goals, social role adaptation, and collaborative task-solving within a multiuser environment. Since these memories have persisted over time, it enabled continuity and personalized interactions and highlights how retrieval-augmented reasoning can be applied to social AI systems.

Frameworks like AutoGen extend the above idea of summaries by providing an environment where multiple agents can communicate, delegate tasks, and specialize in roles [25]. Each agent maintains a local memory, but the framework enforces controlled memory sharing and coordination protocols, thus allowing problem-solving behaviors

that rely on distributed knowledge. Similarly, AgentVerse focuses on a larger scale multi-agent ecosystem where coordination, social dynamics, and communication are central [26]. These platforms illustrate how shared memory channels together with structured agent roles can provide orchestration of team-level cognition, highlighting the need for both runtime memory management and policy enforcement.

Gao et al. (2024) also explored the idea of explicit memory sharing by integrating multiple agents using mechanisms for storing, filtering, and retrieving memories [27]. Their experiments show that shared memory pools not only increase diversity in responses but also enhance capabilities of agents in complex reasoning tasks. Rezazadeh et al. (2025) formalized these ideas with a dynamic bipartite-graph memory representation, encoding user-agent-resource permissions and distinguishing between private and shared memory tiers [28]. This approach demonstrates how access control and time-evolving permissions can be integrated into multi-agent memory fabric, ensuring both privacy and utility.

Transformer-based architectures such as the Shared Recurrent Memory Transformer provide a complementary technical perspective [29]. By pooling and broadcasting working memories across agents, SRMT enables agents to maintain coherent shared knowledge while performing lifelong learning tasks such as pathfinding and multi-agent coordination. Engineering-focused studies, including Mem0 and MIRIX (2025), illustrate the practical considerations of implementing these memory systems at scale, addressing challenges like structured persistence, indexing, sharding, latency, and multi-agent memory orchestration [30]. All these architectures collectively highlight both the potential and the technical requirements of generative agents with memory fabric. Memory fabric would be a combination of memory storage, retrieval, reflection, and controlled sharing to support collaborative intelligence.

3.4 Privacy, protocols, and governance for shared memories

Privacy, security, and governance are obvious critical considerations when designing multi-user memory fabric. A combined multi-agent system paradigm highlights the need for decentralized memory architectures that allow agents to share knowledge without the need for a centralized sensitive data store [32]. These approaches leverage unified learning from multiple agents and differential privacy (DP) techniques to ensure that information derived from one user or agent cannot be directly reconstructed or leaked to others. While learning focuses on model parameter updates, combined memory sharing introduces unique privacy challenges because raw memory such as dialogue histories or task-specific knowledge may itself contain sensitive content. These challenges are analogous to the challenges in preserving memories in a multi-user environment within memory fabric. This necessitates specialized privacy-preserving protocols for memory exchange, as noted in Masoor et al.'s Secure Agent Memory Exchange Protocol (SAMEP) (2025) [31]. The author suggests a protocol which enables semantically searchable, encrypted memory that can be shared among agents, including the ability to search over encrypted indices without compromising privacy.

Contextual privacy is another crucial consideration where the memory fabric must account not only for what is stored, but also for when and how it can be shared. This includes implementing lifecycle controls, user consent management, and selective exposure of memory content based on user-defined policies. Differential privacy techniques

provide a mathematical framework to limit information leakage from aggregated memory stores while quantifying the trade-off between utility and privacy. Although in practice, designing privacy-aware shared memories requires balancing accessibility with strict access control mechanisms, including encryption, role-based permissions, and audit trails.

Studies on user perceptions have suggested that individuals often have weak understanding of how their information is shared in collaborative AI systems. This underscores the importance of a more transparent user interface, consent management, and clearly communicated memory-sharing policies to maintain trust and accountability. Governance mechanisms are also critical to ensure consistency and reliability in multi-user memory fabric. Provenance-aware models, such as CRDTs and knowledge graph provenance frameworks allow systems to track the origin, modification history, and confidence of memory entries, supporting conflict resolution [33, 35]. These mechanisms are essential in team or multi-agent settings, where multiple users contribute to a shared memory space within memory fabric, and inconsistencies or conflicts could compromise task performance or trust.

Overall, privacy, protocols, and governance within memory fabric require an integrated approach combining cryptographic techniques, access control, contextual lifecycle management, and provenance tracking. This ensures that shared memories remain useful for collaboration while protecting individual privacy and supporting transparency, traceability, and regulatory compliance in a multi-user AI system.

3.5 Provenance, consistency, and graph-structured memory

Provenance and consistency mechanisms are essential for maintaining reliability and trustworthiness in shared memory systems, particularly in multi-user and multi-agent contexts. A good agent must be stateful but also reliable in its response, maintaining its consistency, which is pivotal and intricate to memory fabric. Conflict-free replicated data types provide a formal foundation for achieving eventual consistency in distributed memory systems, ensuring that memory writes are reconciled and merged agnostic to the order of operations [33]. While CRDTs are primarily applied to structured data, their principles offer guidance for designing memory fabric where multiple users or agents may concurrently update shared conversational or task-related memories. However, mapping CRDT-style format to semantically dialogue content is non-trivial, as textual knowledge can be ambiguous and dependent on the context [34].

Provenance-aware knowledge that keeps track of origins of memory enhances the reliability and auditability of shared memories. Surveys and frameworks for provenance in knowledge graphs emphasize the importance of recording user metadata about who created or modified a memory, when changes occurred, and the associated confidence levels [35]. Such metadata helps with conflict resolution, traceability, and accountability. Full traceability of knowledge graphs demonstrates practical implementations for triple-level provenance tracking and change monitoring, providing concrete methods for maintaining a transparent memory history. Figure 4 shows the provenance-aware memory fabric graph. KnowWhereGraph offers large-scale structured knowledge graphs that integrate schema design, versioning, and provenance management, providing design patterns directly applicable to shared conversational memory fabric.

Graph-structured memory is particularly useful in a multi-agent setting, as it allows storing memories represented as nodes and edges encoding relationships, or user-agent-resource interactions [36]. By traversing the graph, systems can model dependencies, track contributions, and share updates efficiently while still supporting queries over relational structures. Techniques such as temporal graph embeddings, relational memory networks, and attention-based traversal provide reasoning over shared memories, thus allowing agents to maintain coherence in collaborative tasks and generate contextually appropriate responses. Together, the combination of provenance tracking, CRDT-inspired consistency mechanisms, and graph-structured memory provides a technical foundation for designing auditable, conflict-resilient, and semantically rich multi-user memory fabric.

4 Benchmarking methods

Following are commonly used benchmarking frameworks to validate the fidelity of answers to facts based on short- and long-term memories. Figure 5 shows the benchmark comparisons for memory-augmented dialogue systems:

1. *Very Long-Term Conversations Benchmark (LoCoMo)*: LoCoMo introduces a dataset and evaluation framework designed to test a model's ability to track long-duration, multi-session dialogue with multiple users and events [23]. The conversations are constructed using structured event graphs and temporal dependencies, simulating the way individuals converse and share histories over time.
- *Key contribution*: Unlike short conversations, this benchmark validates whether models can retrieve information shared way earlier based on the current context and reason with it.

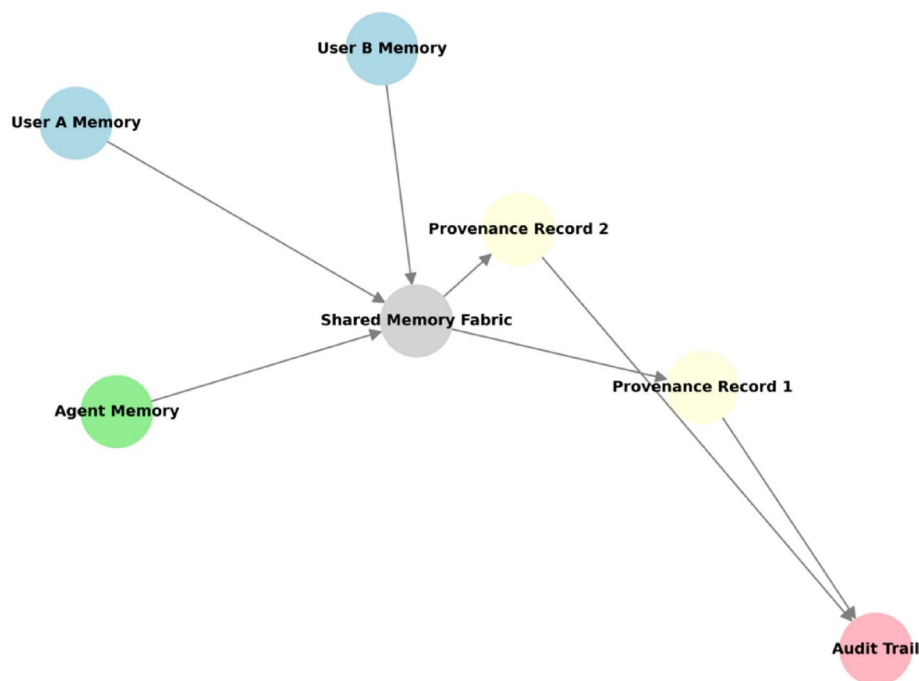


Fig. 4 Provenance-aware memory fabric graph of shared memory systems in multi-user environment

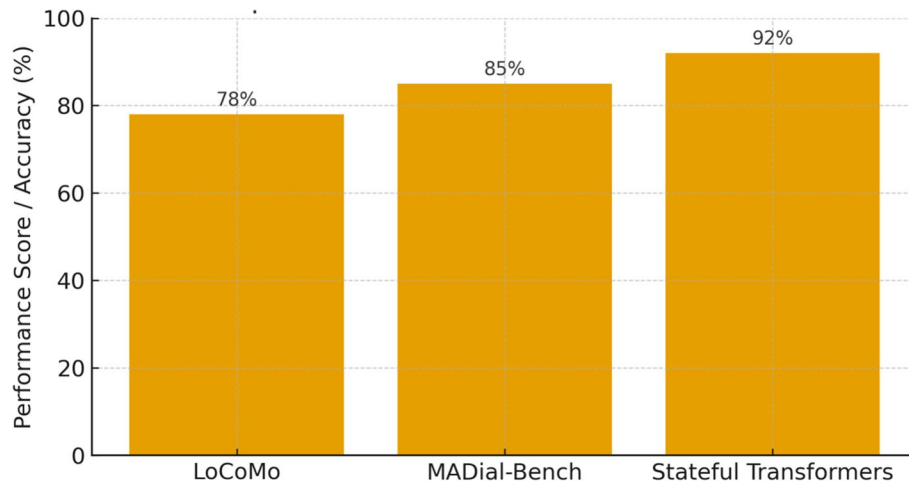


Fig. 5 Comparative benchmark evaluation of accuracy in memory-augmented dialogue systems

- *Findings:* Even LLMs with long context windows or retrieval augmentation struggle with coherence and continuity over long time frames. Thus, human evaluation is still required.

Relevance: Event graphs mirror the structure of memory fabric with provenance and temporally based updates. LoCoMo functions as a testbed for memory correctness, temporal linking, and identity continuity - challenges that amplify in a multi-user environment.2. *Benchmark for Memory-Augmented Dialogue Systems (MADial-Bench):* MADial-Bench was developed to evaluate dialogue systems enhanced with external or persistent memory [22]. Unlike traditional metrics focused on response quality alone, this benchmark measures a model's ability to retrieve past facts, correlate entities across different sessions with minimal hallucinations by constantly updating its memory.

- *Key contribution:* The MADial benchmark introduces dimensions tailored to measure long term memory correctness across different architectures.
- *Findings:* Updating long term memories is harder than retrieving them as models keep both current and outdated information. For example, if a user changes their preference, the model sometimes retrieves outdated information based on past preferences.

Relevance to multi-agent shared memory: Provides a measurable way to assess correctness vs. leakage in memory retrieval. Highlights how memory failures aren't just omission errors — they can also be privacy violations or relevance mismatches.3. *Stateful Memory-Augmented Transformers:* A third class of benchmarks comes from models like MemBART and similar stateful transformer variants, which maintain long-term memories using memory modules rather than regenerating context at every turn. These architectures introduce mechanisms for state persistence across sessions or conversations. Though primarily tested on single-user dialogue settings, they establish architectural baselines for how memory slots are organized, when retrieval is triggered, and how overtime context is merged.

- *Key Contributions:* This benchmark tracks incremental state storage, low-latency retrieval, and efficient indexing without the need to perform full context replay by storing selective history.
- *Findings:* Long range tasks were observed to be more efficient by caching relevant memory. This benchmark motivated software engineers' best practices on integrating memory into deployed systems.
- *Relevance:* These approaches inform the engineering substrate of memory fabric, especially around storage efficiency, write/read policies, and time-based eviction.

5 Discussion

Memory-augmented neural networks have evolved significantly over the past decade, and a comparative analysis reveals their unique strengths and limitations. Neural Turing Machines (NTMs) were among the first architectures to demonstrate that neural networks could learn algorithmic behaviors by using an external, differentiable memory. NTMs can store and retrieve information across long sequences, making them suitable for tasks such as sequence copying, sorting among others. However, applicability to practical applications is limited due to instability during training and inefficient memory addressing, which can lead to difficulty in scaling to larger tasks. Differentiable Neural Computers (DNCs) are built on the foundations of NTMs by the introduction of structured memory management mechanisms, including dynamic memory allocation and temporal linkage systems. These innovations allow DNCs to follow sequences or graph-like structures more reliably, enabling them to solve more complex reasoning problems such as linked list traversal, pathfinding, or multi-step algorithmic tasks. Despite these improvements, DNCs remain computationally intensive and are primarily used for research purposes rather than large-scale deployment.

Key-Value Memory Networks (KV-MemNNs) offer a different approach by organizing memory into discrete key-value pairs, which separates the addressing mechanism from the stored content. This design allows for efficient content-based retrieval, multi-hop reasoning, and straightforward integration with question-answering systems. KV-MemNNs have proven effective in tasks requiring semantic matching and structured information retrieval but are less general-purpose than NTMs or DNCs, and they often rely on predefined memory slots rather than dynamic learning of memory structures. Popular modern large language models (LLMs), in contrast, incorporate memory implicitly through attention mechanisms and, when combined with retrieval-augmented generation, effectively emulate the behavior of memory-augmented networks that can easily be scaled. Attention mechanisms serve as a form of short-term working memory, while external vector databases act as long-term memory, storing embeddings of past interactions or relevant knowledge. Compared to NTMs, DNCs, or KV-MemNNs, LLMs benefit from vast pre-training, distributed architectures, and the ability to generalize across diverse language tasks, although they still face challenges related to memory efficiency, retrieval accuracy, and long-term context retention.

While a wide range of prior systems have explored persistent memory, shared retrieval, or multi-agent coordination, the notion of memory fabric introduced in this review is conceptually distinct in how it elevates shared memory from an auxiliary system component to a first-class, governed substrate of collaboration. Existing shared retrieval stores primarily provide passive similarity-based access to historical embeddings, and

Table 2 Research directions for memory fabric

Direction	Problem	Proposed Idea	Evaluation approach
Formal semantic CRDTs for natural-language memories	CRDTs ensure convergence only for formal operations; natural language is semantically ambiguous	Defining canonical operations (assert, retract, reattribute, link) with commutative merge semantics; attach provenance metadata	Simulated conflicting edits and human reconciliation; connect to KG triple semantics
Access-asymmetric dynamic bipartite permission graphs	Permissions evolve over time; memories must respect time-varying policies	Formalize time-evolving access graphs and incremental retrieval-time enforcement algorithms	Benchmark latency and policy correctness under dynamic sharing scenarios
Scalable graph-embedding hybrids for memory fabric	Graph-structured memories scale poorly with naive indexing	Hybrid retrieval: embedding-based candidate narrowing + graph-join verification for consistency	Measure precision/recall, query latency, and storage overhead
Quantifying privacy-utility trade-offs	Differential privacy (DP) or encryption degrades usefulness in unpredictable ways	Controlled sharing with varied DP budgets / selective redaction	Evaluate downstream task success and satisfaction (LoCoMo, MADial-Bench)
Provenance-driven conflict resolution	Contradictory memories (e.g., who paid the bill?) lack resolution strategies	Use provenance, confidence scores, recency weighting, and majority-voice heuristics	Assess human trust and error rates on reconciled outputs
User-facing UIs & sandboxing	Users need visibility and trust in shared edits; current tools lack team-scale affordances	Extend sandbox UIs for collaborative editing, merge approval, and role-based controls	Measure trust, error rates, and collaboration efficiency in user studies
Evaluation suites for social fairness	Risk amplifying majority voices, silencing minorities with memory fabric	Benchmarks to quantify dominance and marginalization, and apply normalization schemes	Audit fairness on synthetic and real-world collaborative datasets
Versioning & migration across model updates	Embedding/index drift across model versions causes memory loss	Versioned stores with embedding alignment and dual lookup strategies	Test robustness in retrieval across updates (Mem0, MemBART)

agent-level orchestration frameworks focus on routing, role assignment, and coordination protocols among agents. In contrast, memory fabric is defined by the explicit unification of persistence, multi-party ownership, access control, provenance tracking, conflict resolution, and lifecycle management within a single architectural abstraction. As synthesized across the reviewed architectures, memory fabric treats shared memory not merely as a backend data store, but as an evolving socio-technical artifact whose correctness, visibility, and accountability must be modeled alongside retrieval and reasoning. This shift reframes shared memory from a systems optimization into a foundational design layer for multi-user and multi-agent conversational AI.

Overall, the evolution from NTMs to DNCs, KV-MemNNs, and modern LLMs describes a trajectory from explicit, differentiable memory mechanisms toward scalable, embedding-based memory retrieval. (Table 2) Each approach has trade-offs: NTMs and DNCs offer fine-grained, algorithmic memory manipulation suitable for structured reasoning; KV-MemNNs provide efficient semantic retrieval; and LLMs achieve massive scale and flexibility, leveraging attention and external memory to support conversational and reasoning tasks at a practical level [37].

5.1 Advantages and limitations

Memory fabric, the process of integrating, linking, and reconciling memories across multiple users or agents, has emerged as a powerful approach in conversational AI. It allows systems to maintain richer context, enable collaboration, and support long-term

personalization. However, like any emerging paradigm, it comes with notable advantages and limitations. Following advantages:

1. *Enhanced Personalization*: Memory fabric allows conversational agents to draw on the historical interactions of a user and related users to deliver responses that are context-aware and personalized. By leveraging aggregated memory, systems can anticipate user preferences, maintain consistent personas, and provide more human-like interactions.
2. *Improved Long-Term Context Maintenance*: Traditional neural architecture often has limited context windows, restricting their ability to remember information over long conversations or multiple sessions. Memory fabric enables continuity by connecting information across time, ensuring that past events or statements are accessible when needed.
3. *Collaborative Knowledge Sharing*: In multi-agent or multi-user systems, memory fabric enables the sharing of relevant information across participants. This facilitates collective reasoning, collaborative problem-solving, and group decision-making, where each agent or user can contribute to and benefit from a shared knowledge pool.
4. *Support for Complex Reasoning*: Structured memory networks and graph-based weaving approaches allow systems to perform higher-order reasoning, such as tracing dependencies, resolving conflicts, or following temporal sequences. This can enhance decision-making and improve the system's ability to answer queries requiring multi-step inference.
5. *Scalability of Knowledge Integration*: Memory fabric provides a mechanism to consolidate disparate sources of information without hardcoding relationships, allowing the system to scale knowledge acquisition as new users, agents, or documents are added.

Despite sufficient literature to support the practical application of memory fabric, below are a few limitations:

1. *Privacy and Security Concerns*: Sharing memory across users or agents raises significant privacy challenges. Sensitive information may be inadvertently exposed if proper access control, anonymization, or encryption is not implemented.
2. *Complexity in Memory Management*: Memory fabric requires sophisticated data structures, addressing mechanisms, and conflict resolution strategies. Designing, training, and maintaining such systems can be computationally intensive and technically challenging.
3. *Potential for Inconsistent or Conflicting Memories*: When memories from multiple sources are merged, conflicts or inconsistencies can arise. Resolving these requires careful provenance tracking and reconciliation policies, which remain an open research problem.
4. *Difficulty in Evaluation*: Measuring the quality and effectiveness of memory fabric is non-trivial. Metrics for relevance, consistency, recall, and personalization must be carefully designed, and standardized benchmarks are still limited.
5. *Scalability Constraints*: While memory fabric can integrate knowledge from multiple sources, large-scale deployments may face performance bottlenecks. Retrieval, update, and conflict-resolution operations become slower as memory size increases, particularly when semantic similarity or graph-based linking is involved.

5.2 Open problems and research directions

Below are prioritized directions for rigorous research and systems work (these are concrete problems ripe for publishable contributions):

6 Conclusion and future recommendations

Designing and implementing memory fabric in AI systems requires careful consideration of both architectural choices and practical deployment constraints. One key recommendation is to adopt a hybrid memory approach combined with short-term memory, such as attention layers within transformer-based models, along with long-term memory storage using vector databases or key-value memory structures. This combination allows conversational agents to maintain continuity across sessions while efficiently retrieving relevant past interactions or external knowledge. Thereby providing enhanced personalization and contextual awareness [12, 13, 28]. Implementing effective memory management strategies such as dynamic allocation, pruning of outdated or low-relevance memories, and indexing for fast retrieval is essential to prevent information overload and ensure that system responses remain accurate and contextually relevant [27, 29].

Another critical consideration is privacy and access control. Since memory fabric often involves sharing knowledge across users or agents, it necessitates encryption, anonymization, and access restriction policies which are vital to safeguarding sensitive user data and complying with legal and ethical standards [16, 21]. Additionally, robust conflict resolution mechanisms should be integrated when there are multiple sources to contribute to the memory fabric. Provenance tracking based on recency and reliability with algorithms for reconciling contradictory information can help maintain consistency and prevent propagation of errors [18, 30].

From a computational standpoint, engineers should focus on scalability and efficiency. Memory retrieval operations, especially those based on semantic similarity or graph traversal, can become computationally expensive as memory size grows. There are techniques such as approximate nearest neighbor search, embedding compression, and hierarchical memory organization that can mitigate performance bottlenecks while preserving retrieval accuracy [27]. Continuous monitoring and evaluation of memory performance is also recommended using metrics such as relevance, recall, consistency, and user satisfaction.

Memory fabric is understood as a merging, reconciliation of memories across agents and users that has moved from a concept to a technically viable solution. This shift is driven by several factors, like hybrid memory architecture that combines embeddings with structured representations, increasingly capable multi-agent frameworks, and benchmarks such as LoCoMo and MADial-Bench that expose privacy challenges in realistic settings.

At the same time, the field has surfaced a number of unresolved gaps that limit current systems. These include the absence of formal semantics for conflicting or overlapping memories, limited methods for evaluating privacy–utility trade-offs, and the lack of interaction paradigms to cultivate user trust. The result is that scope for memory fabric exists, but the governance, safety, and multi-party reliability layers remain underdeveloped. The research and engineering agenda outlined above translates these gaps into a set of concrete directions.

In sum, the groundwork for shared, multi-user memory architectures is already visible across dialogue systems, retrieval-augmented models, and long-term agent frameworks. What remains is to systematize these elements into interoperable fabrics that are not only technically coherent but also socially accountable. Pursuing the targeted projects identified would accelerate progress toward that goal and generate both high-impact academic contributions and deployable real-world infrastructure.

Author contributions

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No datasets were generated or analysed during the current study.

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